

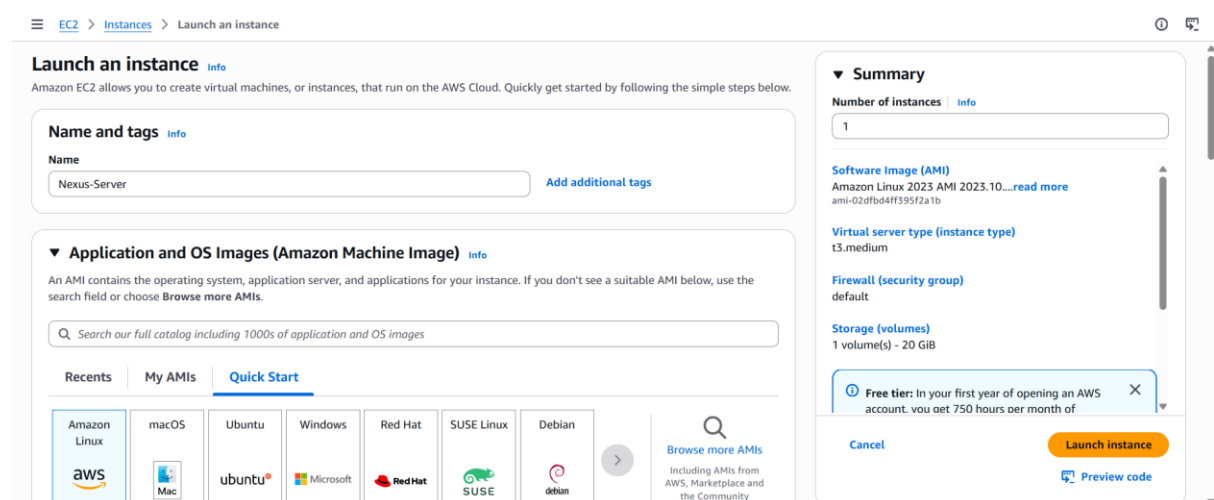
Set-Up of Nexus Repository Server

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to set up a **Nexus Repository Manager** on an AWS EC2 instance. First, an Amazon Linux EC2 instance is launched with required ports (22, 80, and 8081). Java is installed, and the Nexus package is downloaded and extracted in the /opt directory. A dedicated **nexus** user is created and given proper permissions. Nexus is configured to run as a service using **systemd**, then enabled and started. Finally, the Nexus web interface is accessed through the browser, and the admin password is retrieved to complete the initial setup.

- Launch EC2 Instance
- Go to AWS EC2 → Launch Instance
- Configuration:
 - Name: Nexus-Server
 - AMI: Amazon Linux
 - Instance Type: t3.medium
 - Storage: 20 GB
 - Key Pair: existing key pair
- Security Group
- Open these ports:
 - SSH → 22
 - HTTP → 80 (optional)
 - Nexus → 8081
 - Launch the instance.



- Connect to the Instance
- Install java on the system
 - `sudo yum install java-1.8.0-amazon-corretto-devel`

```
[ec2-user@ip-172-31-9-104 /]$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment Corretto-8.482.08.1 (build 1.8.0_482-b08)
OpenJDK 64-Bit Server VM Corretto-8.482.08.1 (build 25.482-b08, mixed mode)
```

- Move to Installation Directory
- Nexus is usually installed in the /opt directory.
 - cd /opt
- [Download Nexus](#)
- Download Nexus repository package:
 - sudo wget https://download.sonatype.com/nexus/3/nexus-3.90.1-01-linux-x86_64.tar.gz

```
[ec2-user@ip-172-31-9-104 /]$ cd /opt
[ec2-user@ip-172-31-9-104 opt]$ sudo wget https://download.sonatype.com/nexus/3/nexus-3.90.1-01-linux-x86_64.tar.gz
--2026-03-14 15:35:16-- https://download.sonatype.com/nexus/3/nexus-3.90.1-01-linux-x86_64.tar.gz
Resolving download.sonatype.com (download.sonatype.com)... 52.8.54.153, 13.56.83.105
Connecting to download.sonatype.com (download.sonatype.com)|52.8.54.153|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.90.1-01-linux-x86_64.tar.gz [following]
--2026-03-14 15:35:16-- https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.90.1-01-linux-x86_64.tar.gz
Resolving cdn.download.sonatype.com (cdn.download.sonatype.com)... 104.18.1.134, 104.18.0.134, 2606:4700::6812:86, ...
Connecting to cdn.download.sonatype.com (cdn.download.sonatype.com)|104.18.1.134|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 465604118 (444M) [application/x-gzip]
Saving to: 'nexus-3.90.1-01-linux-x86_64.tar.gz'

nexus-3.90.1-01-linux-x86_64.tar.gz 100%[=====] 444.03M 153MB/s in 2.9s

2026-03-14 15:35:19 (153 MB/s) - 'nexus-3.90.1-01-linux-x86_64.tar.gz' saved [465604118/465604118]

[ec2-user@ip-172-31-9-104 opt]$
```

- Extract Nexus
 - sudo tar -xvzf nexus-3.90.1-01-linux-x86_64.tar.gz
- Rename Nexus Folder
- Rename the version folder to nexus for easier management.
 - sudo mv /opt/nexus-3.90.1-01 /opt/nexus
- Now the directory structure looks like:
 - /opt/nexus
 - /opt/sonatype-work

```
[ec2-user@ip-172-31-9-104 opt]$ sudo mv /opt/nexus-3.90.1-01 /opt/nexus
[ec2-user@ip-172-31-9-104 opt]$ ls
aws  nexus  nexus-3.90.1-01-linux-x86_64.tar.gz  sonatype-work
[ec2-user@ip-172-31-9-104 opt]$
```

- Create Nexus User
- Create a dedicated user:
 - sudo adduser nexus
- Change Ownership
- Give the Nexus user permission for the directories.
 - sudo chown -R nexus:nexus /opt/nexus
 - sudo chown -R nexus:nexus /opt/sonatype-work
- Verify:
 - ls -l /opt
- Expected output should show nexus nexus ownership.

```
[ec2-user@ip-172-31-9-104 bin]$ ls -l /opt
total 454692
drwxr-xr-x. 4 root root      33 Feb 27 08:23 aws
drwxr-xr-x. 6 nexus nexus    94 Mar 14 15:38 nexus
-rw-r--r--. 1 root root 465604118 Mar  6 20:58 nexus-3.90.1-01-linux-x86_64.tar.gz
drwxr-xr-x. 3 nexus nexus    20 Mar  6 19:02 sonatype-work
[ec2-user@ip-172-31-9-104 bin]$
```

- Configure Nexus to Run as Nexus User
- Navigate to the Nexus bin directory:
 - `cd /opt/nexus/bin`
- Open the configuration file:
 - `sudo vi nexus.rc`
- Find this line:
 - `#run_as_user=""`
- Change it to:
 - `run_as_user="nexus"`
- Save and exit (Esc → :wq → Enter).

```
run_as_user="nexus"
```

- Create Nexus Systemd Service
- Create a service file:
 - `sudo vi /etc/systemd/system/nexus.service`
- Add this configuration:

[Unit]

Description=nexus service

After=network.target

[Service]

Type=forking

LimitNOFILE=65536

User=nexus

Group=nexus

ExecStart=/opt/nexus/bin/nexus start

ExecStop=/opt/nexus/bin/nexus stop

User=nexus

Restart=on-abort

[Install]

WantedBy=multi-user.target

- Save and exit.

```
[Unit]
Description=nexus service
After=network.target
[Service]
Type=forking
LimitNOFILE=65536
User=nexus
Group=nexus
ExecStart=/opt/nexus/bin/nexus start
ExecStop=/opt/nexus/bin/nexus stop
User=nexus
Restart=on-abort
[Install]
WantedBy=multi-user.target
```

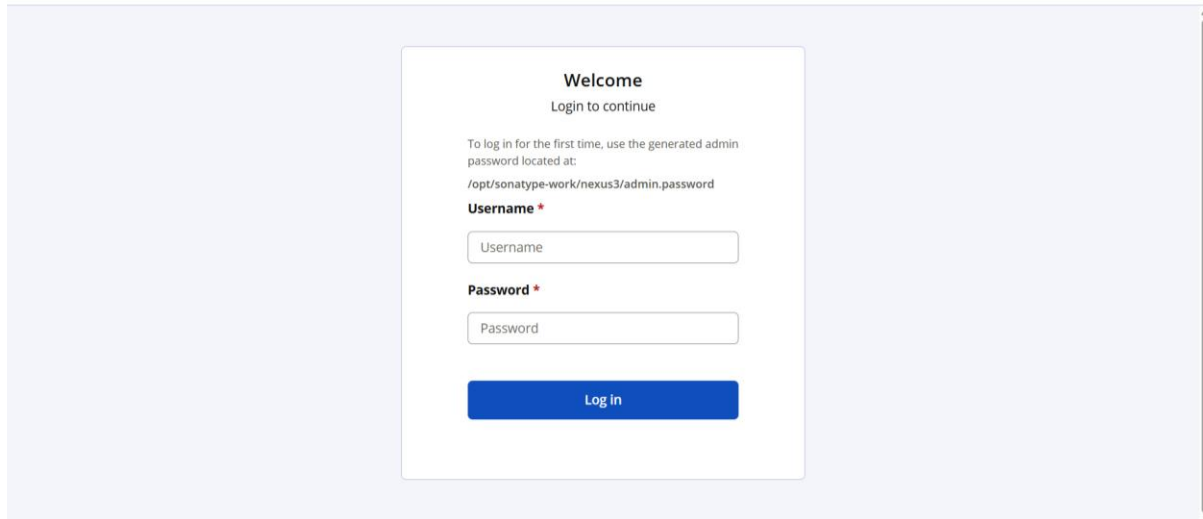
14,26 All

- Reload Systemd
- `sudo systemctl daemon-reload`
- Enable Nexus at Boot
- `sudo systemctl enable nexus`
- Start Nexus
- `sudo systemctl start nexus`
- Check status:
- `sudo systemctl status nexus`
- Expected output:
- Active: active (running)

```
[ec2-user@ip-172-31-9-104 bin]$ sudo vi nexus.rc
[ec2-user@ip-172-31-9-104 bin]$ sudo vi /etc/systemd/system/nexus.service
[ec2-user@ip-172-31-9-104 bin]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-9-104 bin]$ sudo systemctl enable nexus
Created symlink /etc/systemd/system/multi-user.target.wants/nexus.service - /etc/systemd/system/nexus.service.
[ec2-user@ip-172-31-9-104 bin]$ sudo systemctl start nexus
[ec2-user@ip-172-31-9-104 bin]$ sudo systemctl status nexus
● nexus.service - nexus service
   Loaded: loaded (/etc/systemd/system/nexus.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-03-14 16:11:23 UTC; 10s ago
     Process: 31773 ExecStart=/opt/nexus/bin/nexus start (code=exited, status=0/SUCCESS)
    Main PID: 32020 (java)
      Tasks: 21 (limit: 4574)
     Memory: 674.0M
        CPU: 21.628s
    CGroup: /system.slice/nexus.service
            └─32020 /opt/nexus/jdk/temurin_21.0.9_10_linux_x86_64/jdk-21.0.9+10/bin/java -server -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m -XX:+Un

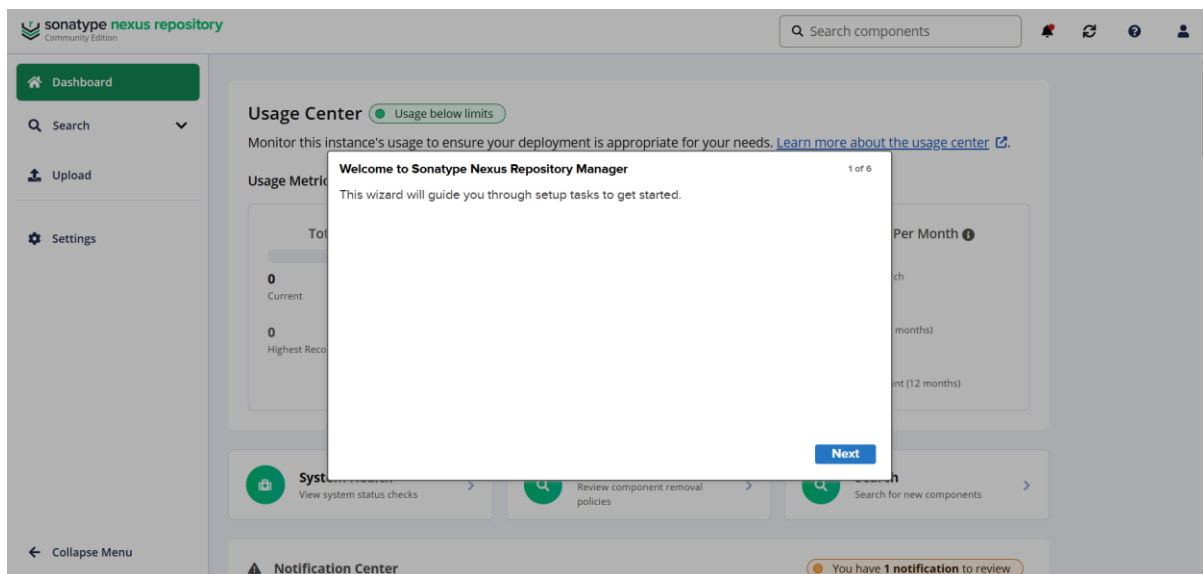
Mar 14 16:11:22 ip-172-31-9-104.ec2.internal systemd[1]: Starting nexus.service - nexus service...
Mar 14 16:11:23 ip-172-31-9-104.ec2.internal nexus[31773]: Starting nexus
Mar 14 16:11:23 ip-172-31-9-104.ec2.internal systemd[1]: Started nexus.service - nexus service.
lines 1-14/14 (END)
```

- Access Nexus Web UI
- Open browser:
- `http://<EC2_PUBLIC_IP>:8081`
- Nexus takes 1–2 minutes to start.
- Once started you will see:
- Nexus Repository Manager UI



The image shows the login page of Sonatype Nexus Repository Manager. At the top, it says "Welcome" and "Login to continue". Below this, it instructs the user to log in for the first time using the generated admin password located at: `/opt/sonatype-work/nexus3/admin.password`. There are two input fields: "Username" and "Password", both with red asterisks indicating they are required. Below the fields is a blue "Log in" button.

- Login to Nexus
- Default username:
 - admin
- Initial password location:
 - `/opt/sonatype-work/nexus3/admin.password`
- Retrieve password:
 - `cat /opt/sonatype-work/nexus3/admin.password`
- Copy the password.



- Configure Access
- Choose:
 - Enable anonymous access
- Click:
 - Next → Finish